THE OHIO STATE UNIVERSITY

CAR STUDENT MOTORSPORTS: BATTERY ACTIVITIES
STANDARD OPERATING PROCEDURE

According to manufacturer's guidelines and required policies within the laboratory and university special precautions must be taken when operating equipment or performing procedures within the laboratory. The following information includes recommendations for operation and any procedures needed in order to operate equipment within the laboratory. This Standard Operating Procedure will be followed along with the requirements of the Chemical Hygiene Plan and Institutional Biosafety Manual.

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1. Purpose

Guidelines and operating procedures for the safe handling and use of batteries and battery testing equipment within the laboratory and student motorsports facility as well as safe storage and disposal of batteries. Guidelines and required policies within the laboratory and university special precautions must be taken when operating equipment or performing procedures within the laboratory. The following information includes recommendations for handling and testing of batteries:

Brief Safety Overview:

- The lab manager is responsible for training employees using the equipment on site. The training should include a discussion of the known and potential hazards of material used in conjunction with material; an explanation of the relevant policies, techniques and procedures including the proper use of personal protective equipment, emergency/spill procedures and containment equipment (engineering controls).
- Limit access to authorized users.
- Require applicable training.

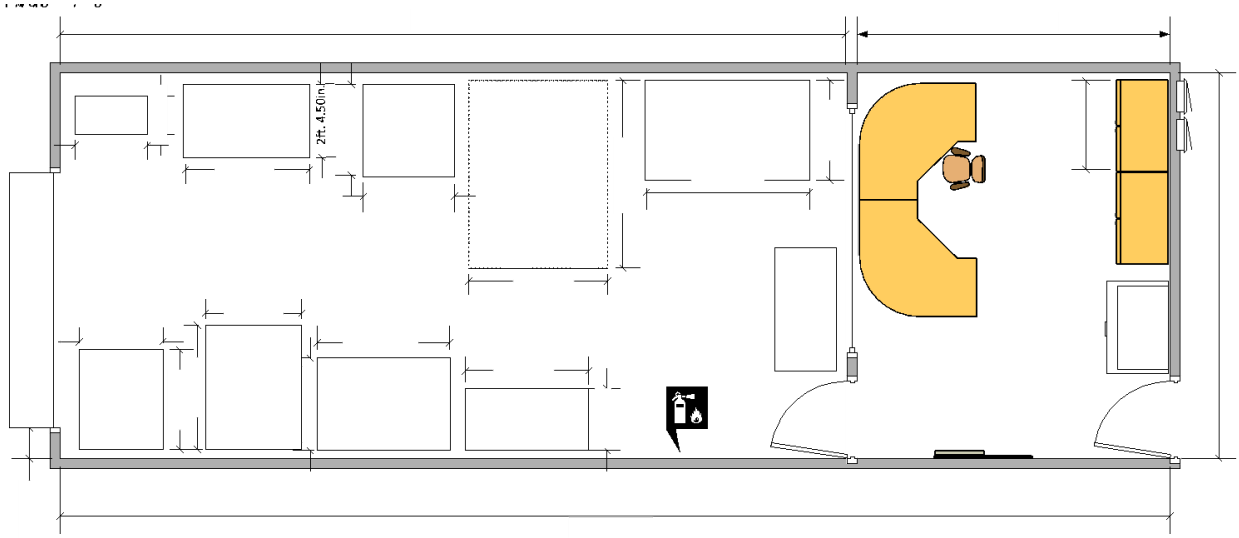
2. Background

Battery Samples: Any cell, module or pack used within the laboratory
Cell: Any single battery sample with a voltage <5V is considered as a cell
Module: A battery sample with multiple cells interconnected. Module can range from 5-60V.
Pack: A battery sample with multiple modules interconnected. Typical packs can range in few hundred volts.

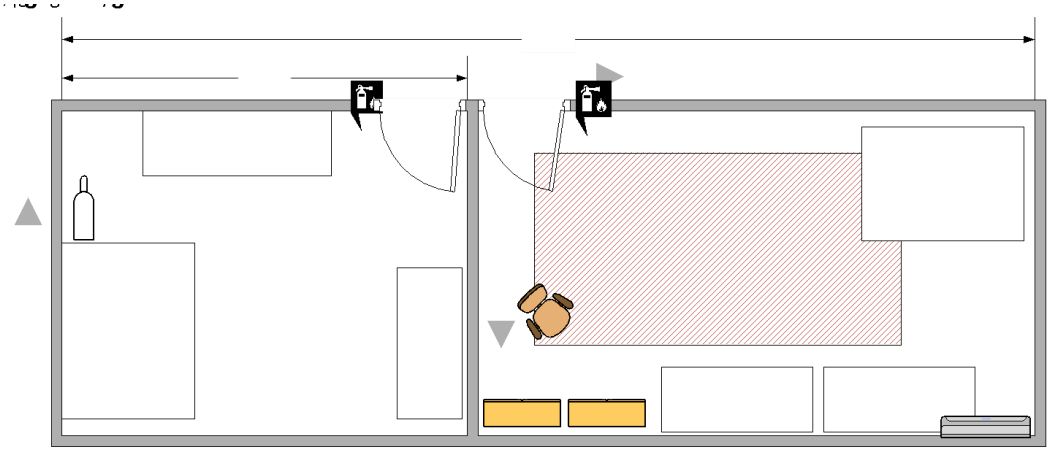
3. Additional Lab Resources:

This document covers the operating procedures pertaining to the battery activities performed by student teams in the motorsports facility at CAR. Additional labs are available in the main CAR building as resources to the students:

- **Low Voltage Lab**
    - Room: 180P, 180L First Floor



- **High Voltage Lab**
  - Room: 180X
- **Pack Fabrication Lab**
  - Room: 180K



Access to the labs in the main CAR building can be obtained from the battery lab manager. Contact details are included at the end of this document.

4. Student Team Representative

Each team should have a single battery lead who ensures all guidelines and operating procedures are being followed. Lab managers should be contacted though the battery lead for the team.

5. Handling:

5.1. General cell/sample:

Hazards and risks can be minimized during handling of samples by following the guidelines below:

- Remove jewelry and accessories such as rings, wristwatches, bracelets , pendants, etc. that could come in contact with battery terminals
- Cell(s) with external physical damage should not be handled without appropriate safety gear and damaged samples should be stored and arranged for disposal.
- Any cell sample that has undergone any physical impact (including falling on the floor) should be stored separately and arranged for disposal and not be used thereafter.
- Cover all metal work surfaces with insulating material to prevent short circuit of samples.
- If cells are not in original packaging and have terminals exposed, arrange in such a way as to prevent potential shorting. Place cells in containers or bags that are non-conductive. Modules and packs should have the terminal covered and not exposed.
- Excessive force should not be used in handling battery samples. Do not pry or force battery samples with tools and/or fixtures.

5.2. Unique (High Voltage) Handling

- Special precaution should be taken when handling samples of high voltage (greater than 55-60V).
- Wear safety glasses and high voltage gloves at all times when handling high voltage modules and battery packs.
- Inspect high voltage gloves for tears, holes, or cracks before using.
- Use dedicated high voltage tools when working on high voltage samples. Contact lab manager for access to tools.
    - Requirements for tools
- When working on high voltage systems including battery packs and modules, there should be at minimum of two people in the room.
- Signage should be in place while working on high voltage samples and access should be restricted to key personnel in the area.

5.3. PPE required:

Following PPE is available in the lab and required depending on the type of test and activity performed within the lab

- Eye Protection (safety goggles)
- High Voltage safety gloves
- Work gloves
- Hand protection (Nitrile gloves)

6. Operation:

Following guidelines should be considered when operating battery samples

6.1. Electrical Testing

Guidelines to be followed while testing battery samples

6.2. Equipment:

6.3. Charging Protocol

- Read and understand the manufacturer's instructions for charging the lithium-ion battery. These instructions should include the recommended charging rate, maximum safe voltage, and other important safety considerations.

- Use a battery charger that is compatible with the lithium-ion battery and follows the manufacturer's recommended charging rate.

- Connect the battery charger to the lithium-ion battery according to the manufacturer's instructions.

- Monitor the voltage and temperature of the lithium-ion battery during the charging process using a battery management system (BMS). The BMS should be set to shut off the charging process if the battery becomes unstable or reaches its maximum safe voltage.

- Do not leave the lithium-ion battery unattended during the charging process.

- Disconnect the battery charger from the lithium-ion battery when the charging process is complete.

6.4. Discharging

- Read and understand the manufacturer's instructions for discharging the lithium-ion battery. These instructions should include the recommended discharging rate, minimum safe voltage, and other important safety considerations.

- Use equipment that is compatible with the lithium-ion battery and follows the manufacturer's recommended discharging rate.

- Connect the equipment to the lithium-ion battery according to the manufacturer's instructions.

- Monitor the voltage and temperature of the lithium-ion battery during the discharging process using a battery management system (BMS). The BMS should be set to shut off the discharging process if the battery becomes unstable or reaches its minimum safe voltage.

- Do not leave the lithium-ion battery unattended during the discharging process.

- Disconnect the equipment from the lithium-ion battery when the discharging process is complete.

6.5. **BMS**

6.6. **Integration (Module/Pack Fabrication)**

The fabrication lab space is utilized to fabricate battery module and pack prototypes and special care and procedures have to be followed to minimize the possibility of hazards. Following are operation procedures to be followed within the fabrication lab space:

- All modules and packs being fabricated must be done so within the lab spaces. Access to the lab is restricted to authorized personnel only.
- At the minimum two people should be present in the area while the welder is actively being used for fabrication.
- No more than 3 people in the room
- Make sure cells, modules and pack terminals remain insulated or covered in order to minimize risk.
- Welding cells poses the risk for short circuit and fire hazards. In the event of any hazard follow emergency procedures detailed in Section 8.
- Welders use compressed gas for pressuring the welding head. When not in use make sure the compressed gas valve is turned off to prevent leakage. Additional compressed gas procedures can be found at: compressed_gas_safety_program.pdf (osu.edu)

7. Storage:

All battery samples must be stored in dedicated spaces with clear labelling indicating batteries when not in use. Follow the following procedures for storage:

- Battery cells should be stored in well-ventilated, dry area normally between 60 and 80 degrees Fahrenheit. Observe manufacturer's minimum and maximum storage temperatures.
- If battery samples being stored are new inventory, prepare a document with the following information and provide to the lab manager for documentation:
    - Manufacturer and name
    - Format (cylindrical, pouch, prismatic)
    - Nominal Voltage
    - Number of cells
    - Project/source of cells
    - MSDS sheet (if available)
- When possible, keep battery samples in original packaging or separated from other cells in separate packaging.
- All terminals must be covered or insulated while under storage to prevent short circuit of the battery.
- All storage areas of batteries should have access to both a class ABC and class D extinguisher.

8. Emergency Procedure:

Handling and use of batteries expose individuals to a certain level of risks: ranging from explosive and fire hazards to shock. Following are procedures to be followed in the event of emergency situations that occur within the battery lab spaces:

8.1. Fire

- Evacuate the lab area and follow procedures detailed in the Building Emergency Action Plan (BEAP) for CAR. BEAP plan can be found on the Safety data binders in the front of the Students Projects Building or the middle of High-Bay Yellow safety board located in CAR.
- Sound the fire alarm to warn others.
- Fires should be put out using the extinguisher only be personnel trained in the use of fire extinguishers.
- The Battery Lab manager should be informed of the event and any details that could help identify the cause of the fire.

8.2. Short Circuit

- Identify if the battery or cell is under an active or if a momentary short has occurred. An active short can lead to a fire if not addressed appropriately.
- If the cause of the external short can be identified such as metal tools, equipment, etc disconnect the cell or isolate the cell immediately.
- If the cell is hot, evacuate the area immediately and notify the Battery Lab manger and Facilities supervisor, as the sample might have to be isolated to prevent further risks.

- Handle the shorted cell with appropriate protective gear.
- Notify the Battery Lab manager of any shorted samples to allow for proper documentation, storage and disposal of the sample.

8.3. Power failure

- During a power failure, one or multiple equipment might lose electrical power and stop working. Risk could be high if the certain equipment stops working during a test.
- If a power failure occurs, identify if any battery samples were under an active test and make sure to disconnect the sample from the test equipment.
- Make sure the battery sample is not under extreme temperatures and make a plan for the sample to be brought back to ambient storage conditions.
- Notify Battery Lab manager of event and steps taken.

9. Equipment Disposal Procedures

- All battery samples no longer being tested or designated for disposal must be stored in the designated storage area until they can be appropriately disposed.
- Follow procedures detailed in section 6 for safe storage of samples.
- Documentation of the sample with details such as, sample type, chemistry, format, status and voltage is required.
- Notify the Battery Lab manager of any samples that require disposal.

10. Contact Information

All laboratory personnel must be familiar with safe handling practices (i.e., training with documentation of training) when working with the equipment in the laboratory spaces, for any reason.

Other questions or concerns, please contact:

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